Work Hour Prediction: Exploratory Data Analysis Report
Group 9

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Abstract

In today's dynamic work environment, understanding the factors that influence employees' weekly working hours is crucial for enhancing productivity and ensuring well-being. This report presents the findings of an exploratory data analysis conducted on the "Work Hour Prediction" dataset obtained from Kaggle. The primary objective of this analysis was to examine how weekly working hours are associated with various demographic and occupational factors. Additionally, the report provides preliminary insights into fitting a predictive model for the dataset and identifies key features that contribute to an optimal model for accurately forecasting individual working hours.

"Time is the most valuable thing a man can spend." – Theophrastus

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1. Introduction

In today's diverse and competitive career landscape, many individuals face significant challenges in balancing their professional and personal lives. The term "work hours" refers to the time people allocate to their paid professional activities within a specific period. Technological advancements, demographic shifts, and social transformations have prompted a reevaluation of the traditional eight-hour workday. Various social, demographic, and job-related factors influence an individual's working hours. Analyzing these factors is essential for enhancing workforce productivity and addressing employees' needs in achieving work-life balance.

This study aims to comprehensively understand the variables associated with an individual's working hours by conducting an exploratory data analysis on the provided dataset.

2. Description of the Question

In recent years, there has been a growing interest in understanding the factors that influence employees' weekly working hours. This interest arises from considerations related to work-life balance, productivity, overall well-being, job satisfaction, and economic implications. Understanding these factors can help organizations optimize workforce management, improve employee retention, and develop policies that promote a healthier work environment.

Hence this report aims to achieve the following:

- Exploring factors influencing an individual's weekly working hours.
- Identifying the type of individuals with the lowest weekly working hours.

Finally, this analysis aims to develop a suitable statistical model to predict an individual's weekly working hours based on their relevant factors.

3. Description of the Dataset

The Work hour prediction data set was taken from Kaggle. It contains 18944 observations with 15 number of variables. One of them is "Hours-per-week" which is the response variable for the study. Here is the description of each variable

Variable	Description
Quantitative Variables	
Age	Age of an individual in years
Fnlwgt	Weight assigned to each individual by the census to make the dataset representative of the entire population.
Education-num	Number of years spent for education
Capital-gain	Capital gain for an individual
Capital-loss	Capital loss for an individual
Hours-per-week	Work hours per week in hours of an individual
Qualitative Variables	
Work-class	Working type of an individual
Education	Highest education qualification

Marital status	Marital Status of an individual
Occupation	Occupation of an individual
Relationship	Relationship type of an individual
Race	Type of the race of an individual
Sex	Type of the sex of an individual
Native country	Native country of an individual
Income	Income level of an individual

Table 1 - Variable Description

4. Data Pre-processing

Missing Values: No missing values were found in the dataset.
Duplicates: No duplicate records were found in the dataset.

Not meaningful values: Since we are analyzing working hours, categories like 'never worked' might not be useful. Therefore, observations in the 'not worked' category have been removed from the work class.
Data entry errors: During exploratory data analysis, we found an unusual category, 'South' in native-country, likely due to a misclassification or an error in data entry, as 'South' does not represent a specific country (It may be a country name starting with 'south'). However, it is the second most frequent category in this variable. Given its significant proportion, we retained it to preserve data structure and distribution.

4.1. Feature Engineering

- Recategorized Variables:

Work Class: 7 categories (without removed one) → 4 categories(private, without pay, government, self-employed)

Prev.	Private	Without-pay	State-gov	Federal-gov	Local-gov	Self-emp-not-inc	Self-emp-inc
New	Private	Without-pay	Government			Self-employed	

Table 2 - Recategorization of 'Work Class' Variable

Education: 16 categories → 4 categories (associate degree holder, degree holder, high school graduate, school level)

Prev.	Assoc-voc	Assoc-acdm	Prof-school	Bachelor's	Masters	Doctorate	Some-college	1st-4th
New	assoc-degree-holder		degree-holder				school-level	
Prev.	5th-6th	7th-8th	9th	10th	11th	12th	Preschool	HS-grad
New	school-level							HS-grad

Table 3 - Recategorization of 'Education'' Variable

Marital-Status: 7 categories → 3 categories (married, separated/widowed, unmarried)

Prev.	Married-civ-spouse	Married-spouse-absent	Married-AF-spouse	Divorced	Separated	Widowed	Never-married
New	married			separated/widowed			unmarried

Table 4 - Recategorization of 'Marital-Status' Variable

Race: 5 categories → 3 categories (white, black, other)

Prev.	White	Black	Amer-Indian-Eskimo	Asian-Pac-Islander	Other
New	White	Black	Other		

Table 5 - Recategorization of 'Race' Variable

Native country: 41 categories → 2 categories (USA, Not USA)

- The dataset was reduced from 18944 to 18881
- Then, the dataset was split into training and test sets. The training set consisted of 15105 observations. Descriptive analysis was conducted using this training set.

5. Main Results of Descriptive Analysis

5.1. Univariate Analysis

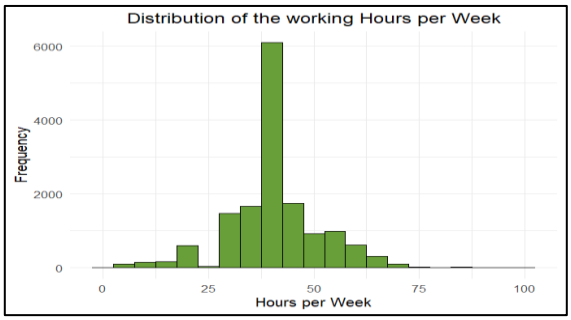

Figure 1 - Distribution of the working Hours per Week

Min	Q1	Median	Mean	Q3	Max
1.00	35.00	39.00	40.12	44.00	102.00

The histogram reveals that working hours per week range from 1 to 102 hours, with a notable peak between 40 and 45 hours. The distribution is approximately symmetric with a right skew, suggesting extremes but it aligns with real-world scenarios that while most individuals work standard hours, some engage in part-time or extended workweeks due to various economic and personal factors [1]

Figure 2 shows the majority of data points near the median, indicating low variability within the central portion of the dataset. A substantial range of outliers on the right side suggests a right-skewed distribution. These outliers align with real-world scenarios. Removing such outliers could lead to a loss of valuable information and may not accurately represent the underlying data distribution

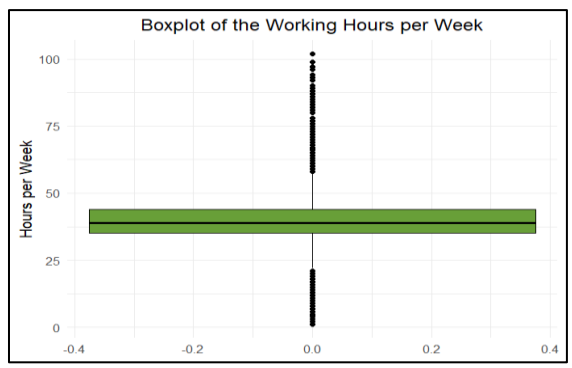

Figure 2 - Boxplot of the Working Hours per Week

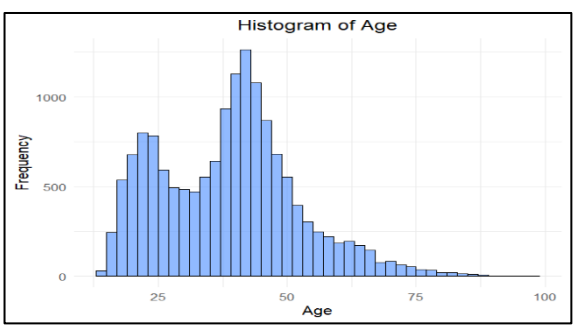

Figure 3 - Histogram of Age

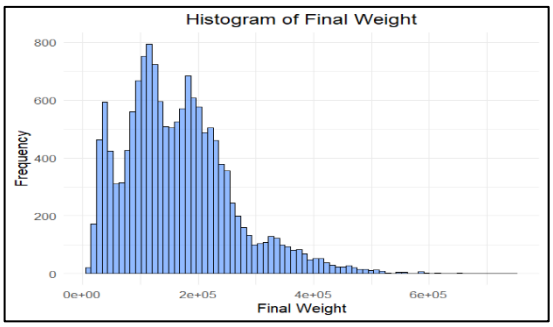

Figure 4 - Histogram of Final Weight

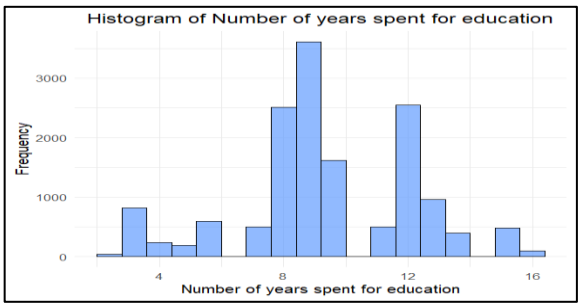

Figure 5 - Histogram of Number of years spent for education

From Figure 3, the age distribution of individuals is positively skewed, with the majority clustered between 20 and 55 years old. From Figure 4 the distribution of Final Weight (the weight assigned to each individual by the census to make the dataset representative of the entire population) is also positively skewed while some outliers can be observed here. And from Figure 5 the distribution of number of years spent for education is positively skewed, with the majority clustered between 8 and 12 years

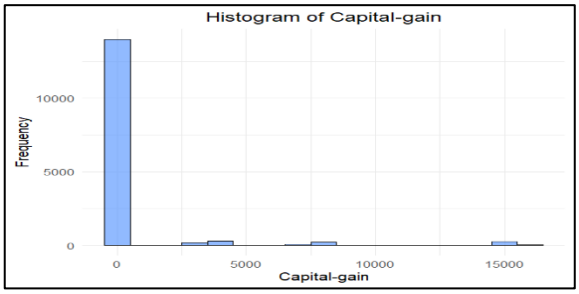

Figure 6 - Histogram of Capital-gain

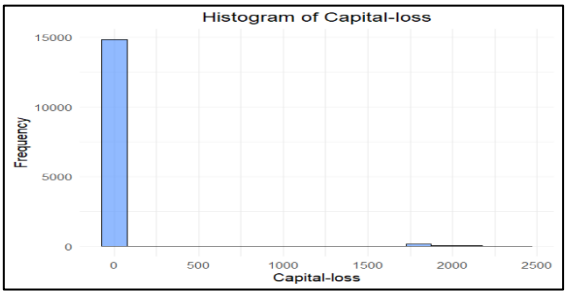

Figure 7 - Histogram of Capital-loss

According to Figure 6 and 7 both capital gain and capital loss variables have a majority of zero values, and the concentration of zero values leads to a highly skewed distribution.

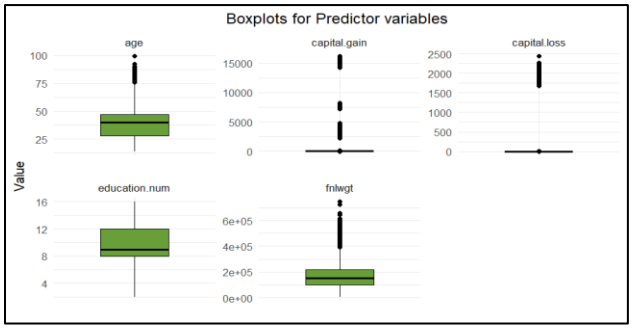

Figure 8 - Boxplots for Predictor variables

Figure 8 shows some variables exhibit outliers, as certain points lie outside the interquartile range. Deciding to remove outliers should not be taken as an easy way. Additionally, the presence of zero values in Capital Gain and Capital Loss variables is common, as many individuals may not have any capital gains or losses in a given period. These zeros are valid observations and should be retained in the dataset.

5.2. Bivariate Analysis

Association between work hours per week vs age category

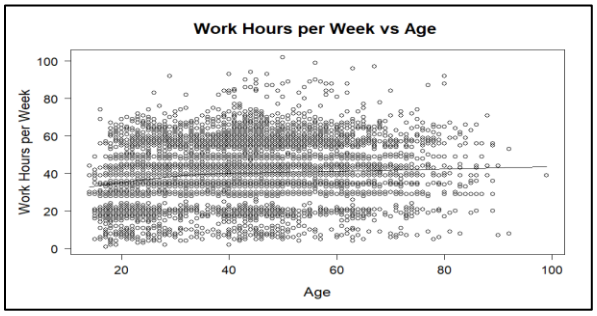

Figure 9 - work hours per week vs age

We can see work hours per week are not strongly corelated with age. There are several 80-year-old people who work more than 80 hours per week which may represent outliers or unique circumstances. There are a lot of people who work around 20hours per week at age 20 or younger, maybe that's because they work part time. We can see a decrease in the number of people working at the age of 80 or more.

Association between work hours per week and Work class

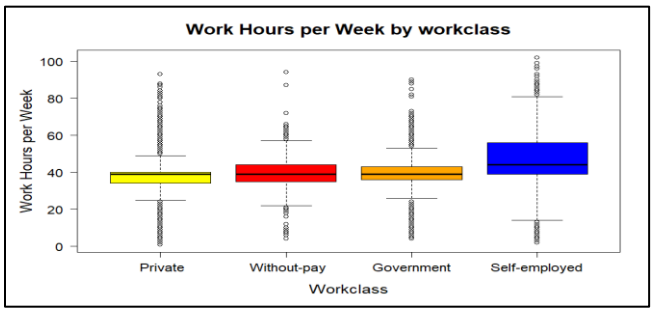

Figure 10 - work hours per week and Work class

Median of private, without-pay and government are close and the median of self-employed is highest, indicating that self-employed individuals tend to work more hours. The "Self-employed" category shows the largest variability in work hours, reflecting the flexibility in schedules in self-employment. The "Private" category has the highest number of outliers, showing a wider range of work hour demands in this sector.

Association between work hour per week and Education level

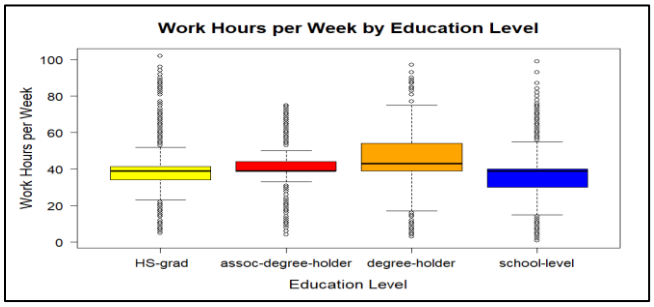

Figure 11 - work hour per week and Education level

By the boxplot we can see median of high school graduates, associate degree holders and school level categories relatively equal but it is high in degree holders this indicates individuals with a degree tend to work more hours compared to other groups. There are outliers in every category we can say there are some individuals at every education level who work significantly more hours than the typical range.

Association between work hours per week and marital status

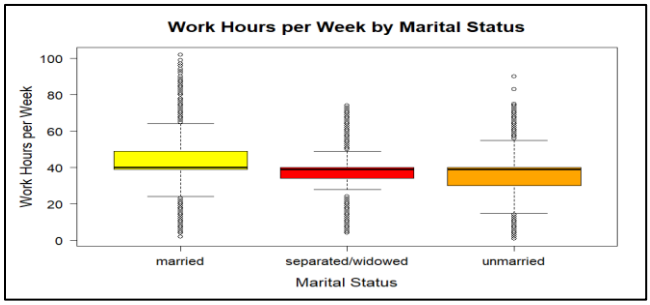

Figure 12 - work hours per week and marital status

By observing the plot, medians of all groups are relatively equal. Married people's working range is higher than other groups. We can see a lot of outliers in this also so we can say there are some individuals from different marital statuses who work significantly more hours than the typical range.

Association between work hour per week and race

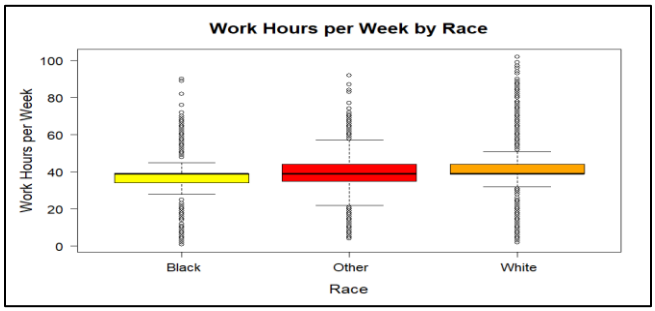

Figure 13 - work hour per week and race

By this plot, we can see how work hours get effect by race so we can see All groups have several outliers and median values of every group is relatively similar. Other racial groups show large variability from all groups. By comparing white and black groups we can say white group individuals work more than black group individual.

Association between work hour per week and Gender

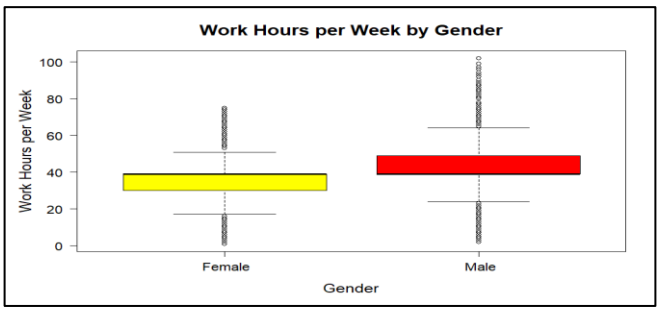

Figure 14 - work hour per week and Gender

By observing this plot we can see males work more hours than females and there are considerbly lot of outliers in male category so we can say there are some males work significantly more hours than the typical range. Median also looks similar. This gender disparity in work hours likely reflects broader societal patterns in the US workforce, including differences in full-time vs part-time employment rates between genders.

Association between work hours per week and native country

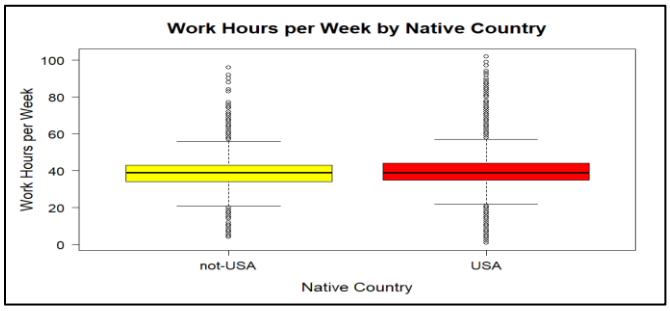

Figure 15 - work hours per week and native country

The box plot comparing work hours between USA natives and non-natives shows significantly similar patterns in both groups, with medians around 40 hours per week. While immigration status often affects employment conditions, this data shows working hours remain consistent regardless of native country. Both groups show outliers.

Association between work hour per week and income

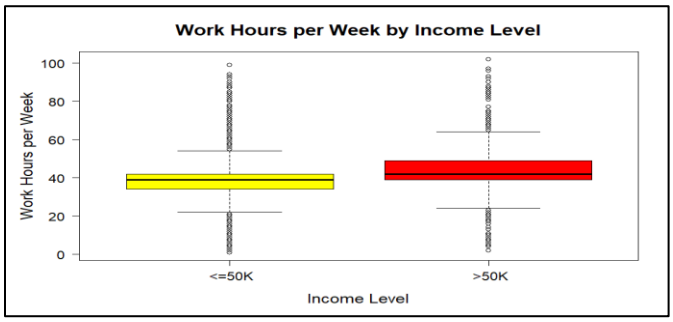

Figure 16 - work hour per week and income

The box plot shows a noticeable difference in work hours between income levels and people earning >50K showing a higher median and larger variability of weekly hours. Both groups have similar lower bonds around 20-25 hours, the >50K group's broader interquartile range indicates higher earners work more hours than low earners in the US labor market. There are lot of outliers in both groups.

Gender vs Income

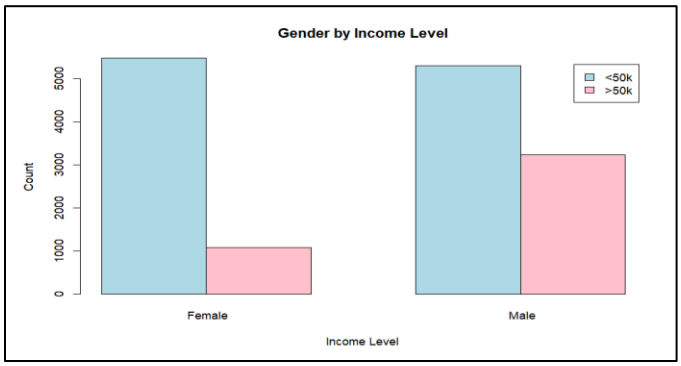

Figure 17 - Plot for Gender vs Income

The bar plot comparing gender and income levels shows noticeable differences. Males show higher representation in the >50k income group compared to females. Larger female workers are in the <50k category, with significantly fewer earning >50k. both male and female representation is looks similar for <50 earners. This suggests gender-based income inequality in the dataset, though further statistical analysis would be needed to confirm these patterns.

5.3. Correlation Analysis

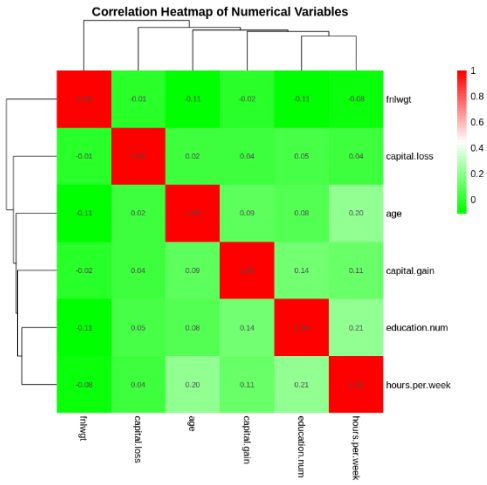

Figure 18

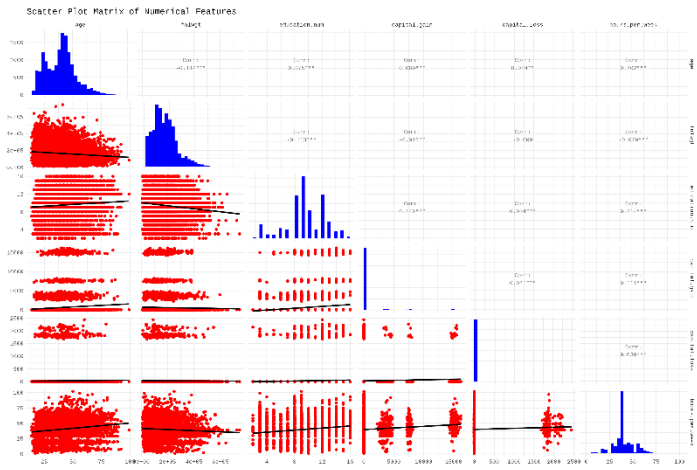

Figure 19

Correlation heat map of numerical variables indicates that there are no significant correlations between numerical variables. Almost all the correlation coefficients between numerical variables are below 0.1 while the highest being 0.21. However, if we look at the distributions of numerical variables some of them ('age', 'fnlwgt', 'capital gain' and 'capital loss' seem to be not normally distributed. Therefore, Spearman's Rank Correlation was considered and the p-values were calculated. (correlation is significant if $p < 0.05$)

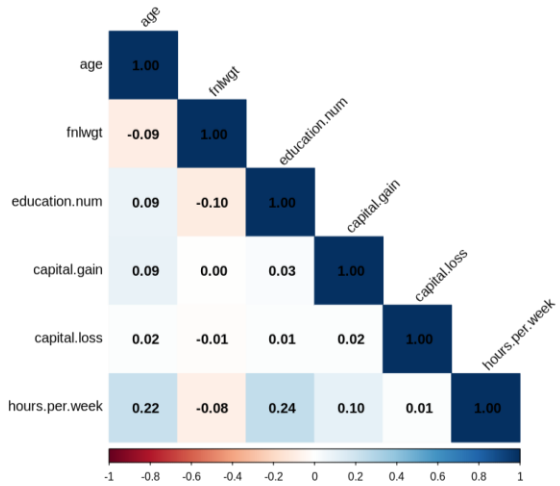

Figure 20

```
                      age       fnlwgt education.num capital.gain capital.loss
age            1.00000000 -0.092071852    0.08799592  0.093045084   0.01583761
fnlwgt        -0.09207185  1.000000000   -0.10249074 -0.003549054  -0.01052138
education.num  0.08799592 -0.102490743    1.00000000  0.029712968   0.01254372
capital.gain   0.09304508 -0.003549054    0.02971297  1.000000000   0.01504387
capital.loss   0.01583761 -0.010521383    0.01254372  0.015043875   1.00000000
hours.per.week 0.22140691 -0.082250301    0.24361966  0.100005981   0.01174967
               hours.per.week
age               0.22140691
fnlwgt           -0.08225030
education.num     0.24361966
capital.gain      0.10000598
capital.loss      0.01174967
hours.per.week    1.00000000
                   age    fnlwgt education.num capital.gain capital.loss
age                 NA 0.0000000  0.0000000000 0.0000000000   0.05160169
fnlwgt        0.00000000        NA  0.0000000000 0.6627260932   0.19599999
education.num 0.00000000 0.0000000           NA 0.0002599028   0.12317409
capital.gain  0.00000000 0.6627261  0.0002599028           NA   0.06447516
capital.loss  0.05160169 0.1960000  0.1231740888 0.0644751555           NA
hours.per.week 0.00000000 0.0000000  0.0000000000 0.0000000000   0.14874106
               hours.per.week
age                 0.0000000
fnlwgt              0.0000000
education.num       0.0000000
capital.gain        0.0000000
capital.loss        0.1487411
hours.per.week             NA
```

Figure 21

Spearman's Rank Correlations have given a location based correlation of the numerical variables. According to the p-values there are significant correlations between several variables. (if the p-value is less than 0.05) However, the Spearman's Rank Correlation Coefficients are in the "Weakly Positive or Negative" range.

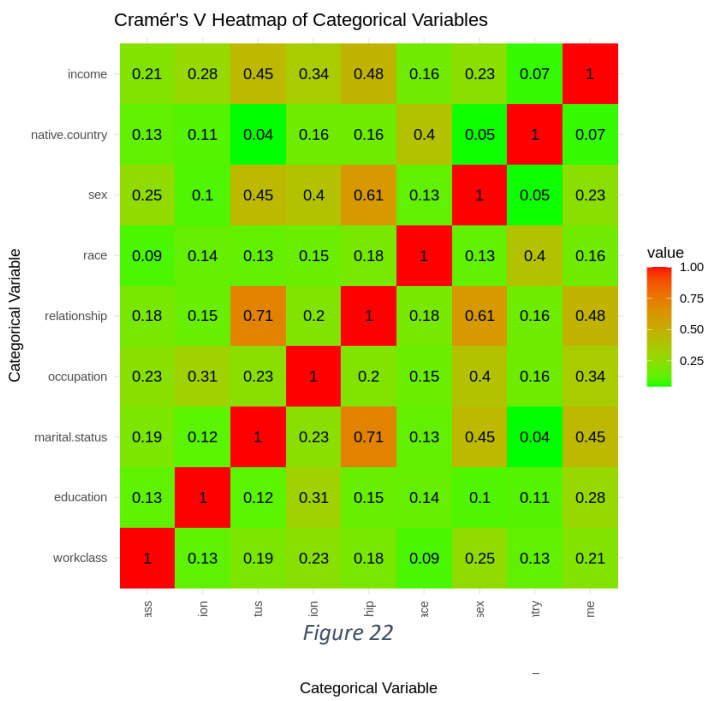

Figure 22

Cramer's V heat map of categorical variables indicates several significant correlations between categorical variables such as 'relationship' and 'marital status'. Furthermore, 'sex' and 'relationship', 'income' and 'relationship', 'income' and 'marital status', 'sex' and 'marital status' indicates moderate correlations.

```
Eigenvalues
                       Dim.1  Dim.2  Dim.3  Dim.4  Dim.5
Variance               3.721  2.235  1.909  1.578  1.378
% of var.             10.057  6.039  5.160  4.265  3.723
Cumulative % of var.  10.057 16.096 21.257 25.521 29.244
```

Figure 23

In such correlations suggestion is to remove redundant variables, merge variables or using multiple correspondence analysis to address those. FAMD was attempted and only 29% of the variance was explained by 5 dimensions. Thus, use of FAMD was considered inadequate.

	GVIF	Df	GVIF^(1/(2*Df))
age	1.477098	1	1.215359
workclass	1.377937	3	1.054884
fnlwgt	1.063608	1	1.031314
education	2.470229	3	1.162669
education.num	2.243541	1	1.497845
marital.status	6.552244	2	1.599917
occupation	2.470452	13	1.035397
relationship	9.367577	5	1.250728
race	1.372179	2	1.082312
sex	1.770184	1	1.330483
capital.gain	1.085497	1	1.041872
capital.loss	1.012404	1	1.006183
native.country	1.302422	1	1.141237
income	1.542422	1	1.241943

Figure 23

Generalized VIF test has been carried out to identify the multi-collinearity that exist between numerical and categorical variables. “hours.per.week” was assumed as the target variable and a linear model was fitted to calculate the GVIF values. There is considerable (>5) multi-collinearity existing between other predictors and “marital.status”, “relationship”.

Thus, suggestion is to merge or remove correlating variables since the data structure was identified too disperse by FAMD and MCA.

In conclusion, there are weak correlations exists among most of the numerical and categorical variables. However, it was discovered that there are moderate to high correlations between some variables, but mostly it was weak correlations. According to the multi-collinearity analysis there were two variables with moderate and high multi collinearity which we have to address during the advanced analysis.

6. Suggestions for Quality Advance Analysis

Through the exploratory data analysis we have identified existence of outliers, multi-collinearity. Thus the following models were suggested.

- Ridge Regression/Lasso Regression - Handles multi-collinearity well. Reduces dimensionality, Outlier effects are minimized because it penalizes large coefficients.
- Random Forest - Naturally handles multi-collinearity and redundant features. Robust to outliers
- XG Boost - Advanced and complex tree based model than Random Forest.